

Proofs: $\rightarrow$ Direct Proofs
Prop: if $x \in \mathbb{Z}$ is odd, then x^2 is also odd.

Proof: Let $x \in \mathbb{Z}$ be odd.

$\Rightarrow$ Then by the defn odd, $\exists k: x = 2k+1$.

$\Rightarrow$ Then $x^2 = (2k+1)^2$

$$= 4k^2 + 4k + 1.$$

$$= 2(2k^2 + 2k) + 1.$$

$$= 2L + 1 \text{ for } L = 2k^2 + 2k.$$

Done!

$\Rightarrow$ Then by definition of odd x^2 is odd. $\square$ QED

Proof 2:

For all $x \in \mathbb{Z}$, $x^2 + x - 6$ is even.

- Satisfy cases $\begin{cases} x \text{ even} \\ x \text{ odd} \end{cases}$

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- 6 is even

- Factor to $(x+3)(x-2)$

- adding 2 odds $\rightarrow$ even

- x odd $\Rightarrow x^2$ odd

Proof by contradiction:

idea: assume $\neg P$ & show a contradiction

want: $P \rightarrow (\neg P) \rightarrow \text{contradiction} \rightarrow \text{QED}$ $\leftarrow$ hang on

ex: Prop $\sqrt{2}$ is not rational

$\rightarrow$ Prop #1 if x is odd, then x^2 is odd

Proof Assume for the sake of contradiction $\sqrt{2}$ is

rational $\exists m, n \in \mathbb{Z} \sqrt{2} = \frac{m}{n}$ where $\frac{m}{n}$ is in reduced form

$$2 = \frac{m^2}{n^2}$$

$$2n^2 = m^2 \Rightarrow m^2 \text{ is even}$$

$\Rightarrow m \text{ is even}$ by Prop #1

$$\Rightarrow m = 2k, \text{ for some } k \in \mathbb{Z}$$

$$p_n^2 = (2k)^2 = 4k^2$$

$$\Rightarrow n^2 = 2k^2$$

$$\hookrightarrow n^2 \text{ is even} \Rightarrow \boxed{n \text{ is even}} \\ (\text{by Prop \#1})$$

But $\frac{m}{n}$ is in reduced form

This is a contradiction $\Leftarrow \Rightarrow \Leftarrow \Rightarrow$

Thus $\sqrt{2}$ cannot be rational ~~QED~~

~~✗~~ not in front of smth $\rightarrow$ Negation

$$\neg (x = 5) \rightarrow x \neq 5$$

$$\neg (x > 0) \rightarrow x \leq 0$$

$$\neg (x > 0 \wedge x \text{ is odd}) \rightarrow \neg (x > 0) \text{ OR } \neg (x \text{ is odd})$$

$$\neg (p \text{ or } q) \rightarrow \neg p \text{ AND } \neg q$$



$$\neg (p \wedge q) = \neg p \vee \neg q$$



De Morgan's Laws